ECI × Token Price Pareto Frontier — Project Spec

Goal

Produce a chart plotting AI models on two axes:

- **X-axis:** price per token (log scale, blended input/output)
- **Y-axis:** Epoch Capabilities Index (ECI) score

Highlight the **Pareto frontier** — the set of models that aren't dominated by any other model on both capability and price — and show how that frontier has shifted over time.

Why

Epoch publishes the ECI (a single general-capability score) and separately publishes price-per-token data, but no official chart combines them. This fills that gap.

Inputs

1. **ECI scores** — from Epoch's public ECI dataset (capability score per model).
2. **Token prices** — from a pricing source such as Artificial Analysis or Epoch's own price dataset (per-model $/M tokens, blended input+output).
3. **Model metadata** — release date, developer/org, open vs closed weights. Needed for filtering and coloring, not for the core axes.

The join key is model name/version. Name normalization between the two sources will be the main data-wrangling headache — flag mismatches rather than silently dropping them.

Output

A single interactive chart (web-based) that shows:

- One point per model.
- Pareto frontier drawn as a stepped line connecting the non-dominated models.
- Hover tooltip with model name, org, release date, ECI, price.
- Clear visual distinction between frontier and non-frontier points.

Must-have features

- Log-scale X-axis (prices span 3+ orders of magnitude).
- Pareto frontier computed and rendered distinctly.

- Legend/key explaining what "frontier" means on this chart.
- Source and data-date footnote on the chart itself.

Nice-to-have features (in priority order)

1. **Color by release quarter** — so the frontier's movement over time is visible at a glance.
2. **Filter toggle:** all models / open-weight only / closed only.
3. **Filter toggle:** include or exclude reasoning models (their per-token price understates total cost per task).
4. **Time-slider** — replay the frontier's evolution month by month.
5. **Export** — PNG and CSV of the joined dataset.

Explicitly out of scope

- Token-count-adjusted "cost per benchmark run" analysis (that's what Epoch's *Price of Progress* paper does; we're doing the simpler per-token view).
- Speed/latency dimension.
- Training compute.
- Any modeling of future trends — this is a snapshot, not a forecast.

Open questions for me to answer before you start

- [ ] Which pricing source? (Artificial Analysis API, Epoch's price CSV, or scrape?)
- [ ] Input/output price blend ratio — 3:1 (Epoch's convention) or something else?
- [ ] Include reasoning models in the main view, or exclude by default?
- [ ] Static site vs notebook vs standalone HTML file?
- [ ] How should the chart handle models with multiple price points over time — latest price only, or all historical points?

Success criterion

I can look at the chart and immediately see (a) which models are currently on the frontier, and (b) whether the frontier has moved down-and-right over the last year.